LAB 7

FUZZING FOR INPUT VALIDATION BUGS

Software Engineering Laboratory Report

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Assigned Repository	L6_L7_59_phone_repair_quote
Application	Phone Repair Quote Parser
Testing Library	Hypothesis

Objective

To apply property-based fuzz testing using the Hypothesis library, identify input-validation bugs in the phone repair quote parser, and improve the robustness of its three functions.

Functions tested

1. **mask_imei()** – masks an IMEI value.
2. **parse_parts_order()** – parses part names and quantities.
3. **parse_model()** – separates a phone brand and model.

1. Methodology and Test Approach

The original application and its tests were run first. Hypothesis generated text and None inputs to exercise edge cases beyond ordinary manually selected examples. The failing tests exposed exceptions caused by unexpected inputs. The application was then updated with input validation and safe handling, and the same fuzz tests were executed again.

2. Bugs Identified Before Fixing

Function	Generated failing input	Observed issue
mask_imei()	None	AttributeError: None has no replace() method
parse_parts_order()	None, empty text, 'x'	AttributeError / ValueError while splitting or converting quantity
parse_model()	None, empty text	AttributeError / ValueError while stripping and splitting

3. Fixes Applied

The application was hardened to validate input types before string operations. The IMEI function returns an empty string for non-string input and safely handles short strings. The parts-order parser returns an empty dictionary for invalid input, skips malformed entries, and accepts only positive integer quantities. The model parser checks that the input contains both a brand and a model before returning a tuple.

4. Final Test Result

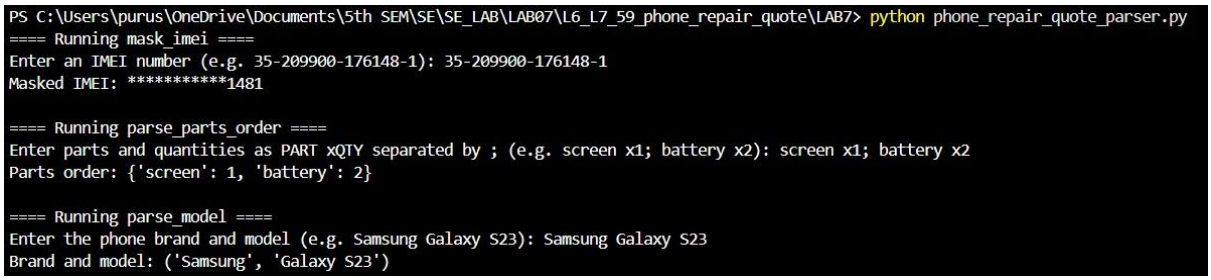
3 tests passed, 0 failed. The final run completed in 18.88 seconds in the recorded terminal output. The test cases covered all three functions using Hypothesis-generated text and None values.

5. Deliverable Source Files

- phone_repair_quote_parser.py – fixed application script
- test_phone_repair_quote_parser.py – updated Hypothesis test script
- This PDF – report with execution evidence and screenshots

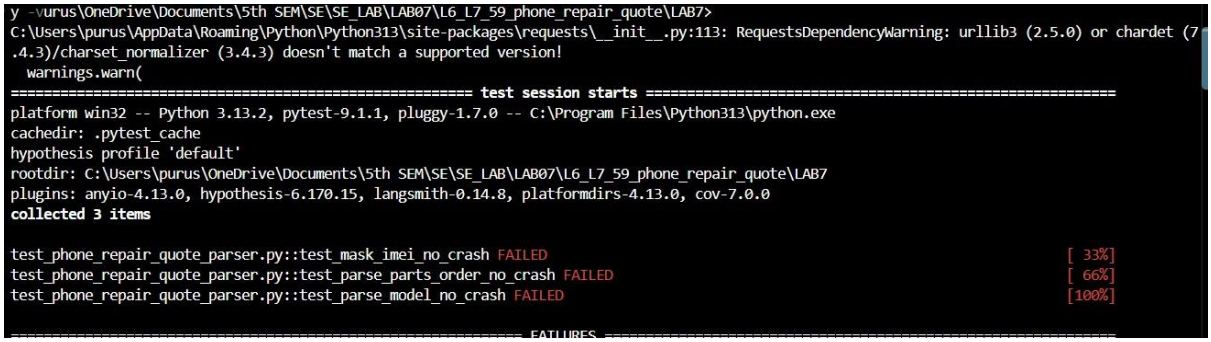
Screenshot 1 – Original application execution

The application was run before fixing the input-validation bugs. The three functions were exercised with ordinary sample inputs.



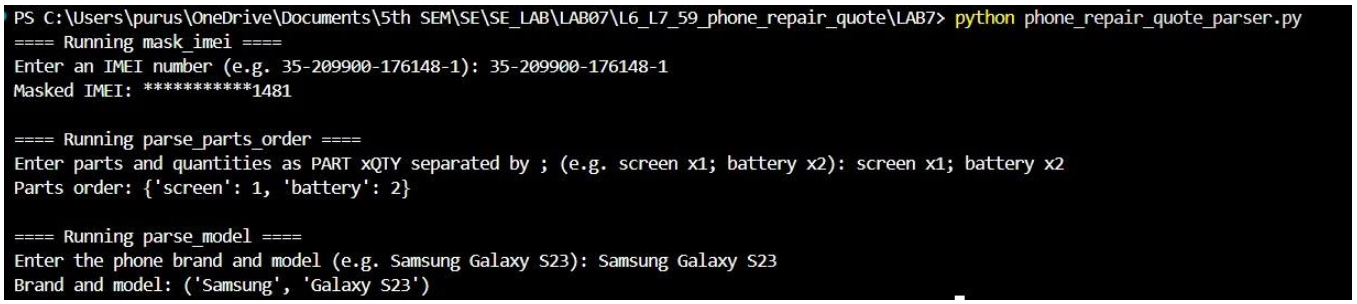
Screenshot 2 – Hypothesis failures before fixing

The original implementation failed all three fuzz tests. Hypothesis exposed None and malformed/empty string inputs that triggered exceptions.



Screenshot 3 – Fixed application execution

After defensive input handling was added, the application completed successfully with the sample IMEI, parts order, and phone model.



Screenshot 4 – Final Hypothesis test results

The updated application passed all three Hypothesis no-crash tests: mask_imei, parse_parts_order, and parse_model.

